where the field-ness is indispensable. And GRW, at least in the simple version presented here, gives us instructions for how wave functions collapse that refers specifically to the probability per particle. This isn't necessarily an insurmountable obstacle—taking simple models that don't quite work and generalizing them until they do is the theoretical physicist's stock-in-trade—but it's a sign that these approaches don't seem to fit naturally with how we currently think about the laws of nature.

GRW delineates the quantum/classical boundary by making spontaneous collapses very rare for individual particles, but very rapid for large collections. An alternative approach would be to make collapse occur whenever the system reached a certain threshold, like a rubber band breaking when it is stretched too far. A well-known example of an attempt along these lines was put forward by mathematical physicist Roger Penrose, best known for his work in general relativity. Penrose's theory uses gravity in a crucial way. He suggests that wave functions spontaneously collapse when they begin to describe macroscopic superpositions in which different components have appreciably different gravitational fields. The criterion of "appreciably different" here turns out to be difficult to specify precisely; single electrons would not collapse no matter how spread-out their wave functions were, while a pointer is large enough to cause collapse as soon as it started evolving into different states.

Most experts in quantum mechanics have not warmed to Penrose's theory, in part because they are skeptical that gravity should have anything to do with the fundamental formulation of quantum mechanics. Surely, they think, we can talk—and did, for most of the history of the subject—about quantum mechanics and wave-function collapse without considering gravity at all.

It's possible that a precise version of Penrose's criterion could be developed in which it is thought of as decoherence in disguise: the gravitational field of an object can be thought of as part of its environment, and if two different components of the wave function have different gravitational fields, they become effectively decohered. Gravity is an extremely weak force, and it will almost always be the case that ordinary electromagnetic interactions will cause decoherence long before gravity would. But the nice thing about gravity is that it's universal (everything has a gravitational field, not everything is electrically charged), so at least this would be a way to guarantee that the wave function would collapse for any